

CHURNGUARD AI

Customer Churn Prediction & Retention Intelligence System

Domain: Artificial Intelligence & Machine Learning

Duration: 2 Days

Language: Python

Environment: Google Colab

Important result note

Run the supplied CHURNGUARD_AI.ipynb before final submission. The notebook calculates the actual evaluation metrics from the dataset. Replace the RUN NOTEBOOK placeholders in this report with those actual values.

1. Problem Statement

Build an end-to-end machine-learning system that analyzes historical telecom customer information and predicts whether a customer is likely to churn.

2. Objective

Analyze customer data; identify patterns; clean and prepare features; train at least two classifiers; compare performance; predict churn probability; create risk categories; and produce business recommendations.

3. Dataset

IBM Telco Customer Churn public sample dataset. Target variable: Churn (Yes/No). Features include demographic, service, contract, payment, tenure and billing information.

4. Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib and Google Colab.

5. Methodology

Raw Data → Data Understanding → Cleaning → Feature Preparation → Model Training → Evaluation → Prediction → Business Insight.

6. Models Tested

Logistic Regression and Random Forest. Day 2 improvement: class-weighted Logistic Regression using `class_weight='balanced'`.

7. Evaluation

Accuracy, Precision, Recall, F1-score, ROC-AUC and Confusion Matrix. F1 is used as the primary selection criterion because churn is the minority class.

8. Business Interpretation

The system can help prioritize customers whose observed characteristics are associated with higher churn probability. High-risk customers can be considered for retention review, service review or targeted offers.

9. Limitations

Predictions are not guarantees. Historical sample data may not represent future behavior. Observational patterns do not prove causation. Thresholds should be calibrated before production use.

10. Future Improvements

Hyperparameter tuning, probability calibration, cost-sensitive thresholding, temporal validation, SHAP explainability, Streamlit deployment and model monitoring.

11. Evaluation Results — Complete After Notebook Run

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK
Random Forest	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK
Logistic Regression (Balanced)	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK	RUN NOTEBOOK

12. Day 2 Improvement

Before F1: RUN NOTEBOOK

After F1: RUN NOTEBOOK

Change: class-weighted Logistic Regression. Explain the observed before/after trade-off using the actual notebook output.

13. Final Results

Dataset records: _____

Selected model: _____

Accuracy: _____

Precision: _____

Recall: _____

F1 Score: _____

ROC-AUC: _____

14. Evidence to Insert

Dataset preview, EDA visualization, model evaluation, confusion matrix, final prediction and final result.